CloudFormation YAML + Manual NAT Gateway Setup

This document contains the CloudFormation template (without NAT Gateway) and step-by-step manual NAT Gateway creation process.

1. CloudFormation YAML

```
AWSTemplateFormatVersion: '2010-09-09'
Description: Dell VPC Setup with Public and Private Subnets

Resources:
  DellVPC:
    Type: AWS::EC2::VPC
    Properties:
      CidrBlock: 10.0.0.0/16
      EnableDnsSupport: true
      EnableDnsHostnames: true

  InternetGateway:
    Type: AWS::EC2::InternetGateway

  AttachGateway:
    Type: AWS::EC2::VPCGatewayAttachment
    Properties:
      VpcId: !Ref DellVPC
      InternetGatewayId: !Ref InternetGateway

  PublicSubnet:
    Type: AWS::EC2::Subnet
    Properties:
      VpcId: !Ref DellVPC
      CidrBlock: 10.0.1.0/24
      MapPublicIpOnLaunch: true

  PrivateSubnet:
    Type: AWS::EC2::Subnet
    Properties:
      VpcId: !Ref DellVPC
      CidrBlock: 10.0.2.0/24

  PublicRouteTable:
    Type: AWS::EC2::RouteTable
    Properties:
      VpcId: !Ref DellVPC

  PublicRoute:
    Type: AWS::EC2::Route
    Properties:
      RouteTableId: !Ref PublicRouteTable
```

```
      DestinationCidrBlock: 0.0.0.0/0
      GatewayId: !Ref InternetGateway

  PrivateRouteTable:
    Type: AWS::EC2::RouteTable
    Properties:
      VpcId: !Ref DellVPC
```

2. Manual NAT Gateway Creation Steps

1. Deploy the CloudFormation stack using the YAML file in AWS CloudFormation.
2. Go to VPC Console > NAT Gateways.
3. Click Create NAT Gateway.
4. Enter Name: dell-nat-gateway.
5. Select Public Subnet: dell-subnet-pub.
6. Choose Connectivity Type: Public.
7. Click Allocate Elastic IP.
8. Click Create NAT Gateway and wait for status Available.
9. Go to Route Tables and select private route table: dell-pvt-rout.
10. Edit Routes and add: 0.0.0.0/0 -> NAT Gateway.
11. Save changes.
12. SSH to Public EC2, then connect to Private EC2 and test internet access.

3. Test Commands

```
ssh -i dell.pem ec2-user@<Public-IP>

ping google.com

sudo yum update -y
```